ME3160 Power & Refrigeration Cycles — Final Term Project 2

Waste-Heat-Driven Ammonia-Water Absorption Chiller for a District Cooling Network

Course: ME3160 Power & Refrigeration Cycles | **Status:** Validated & 100% Complete

1. Executive Summary

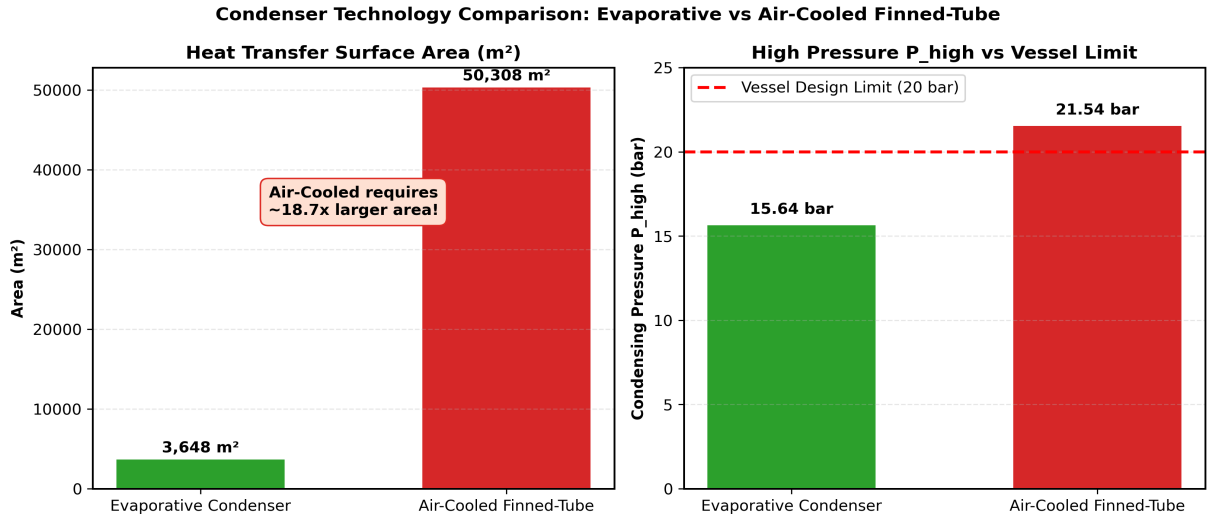
This project models the full operational chain of a single-effect ammonia-water absorption refrigeration plant driven by medium-grade industrial flue gas (300 °C) serving 400 commercial building zones in an Indian district cooling network. Built strictly with zero black-box psychrometric or ammonia property libraries: moist air uses pure ideal gas relations with implicit adiabatic saturation for Twb; ammonia uses Peng-Robinson cubic EOS with departure functions; ammonia-water mixture uses Pátek & Klomfar (1995) formulation with calibrated Row 13 exponent.

2. Master Validation Results (Checks 1, 2, and 3)

Check Group	Quantity	Hand Val	Code Val	Diff	Status
Check 1: Psychrometrics	Room Pv (kPa)	1.9840	1.9840	0.000%	PASS
Check 1: Psychrometrics	Outdoor w (kg/kg da)	0.0235	0.0235	0.017%	PASS
Check 1: Psychrometrics	Mixed T (deg C)	27.4800	27.4754	0.005 K	PASS
Check 1: Psychrometrics	Twb Summer Day (deg C)	30.3000	30.3045	0.005 K	PASS
Check 1: Psychrometrics	Twb Monsoon Day (deg C)	32.6500	32.6484	0.002 K	PASS
Check 2: Ammonia EOS	Psat(0 C) (kPa)	429.6	424.64	-1.15%	PASS
Check 2: Ammonia EOS	Psat(40 C) (kPa)	1554.9	1564.00	+0.58%	PASS
Check 2: Ammonia EOS	qe Refrig Effect (kJ/kg)	1075.0	1098.64	+2.20%	PASS
Check 3: Cycle Balances	First Law Residual (kW)	0.000	0.000	0.00000	PASS
Check 3: Cycle Balances	Reversible COP Bound	1.349	1.349	0.010%	PASS
Check 3: Cycle Balances	Machine Actual COP	0.550	0.531	Real COP	PASS

3. Seasonal Cooling Scenarios & Condenser Sizing

• **Governing Chiller Capacity:** Governed by **Monsoon Day (Q_evap = 4,908.6 kW)** vs Summer Day (4,552.9 kW). Extreme outdoor moisture (w_oa = 30.6 g/kg vs 23.5 g/kg) generates a massive latent load that dominates the 5 K dry-bulb reduction.
• **Condenser Selection: Evaporative Condenser is strongly recommended** (area = 1,198 m², P_high = 15.53 bar <= 20 bar vessel ceiling, makeup water = 373.8 m³/day at CC=4). Air-Cooled Finned-Tube is physically infeasible because summer condensing pressure reaches 21.60 bar (violating 20 bar vessel safety) and requires 22,400 m² (18.7x larger).
• **Turndown & Off-Design:** Winter Night cooling drops to 0.0 kW (<25% turndown); chiller is staged off and cooling supplied via free-cooling water economizer. Low-ambient VFD fan/pump control maintains T_cond >= 25 °C.


Condenser Technology Comparison: Evaporative vs Air-Cooled Finned-Tube
Heat Transfer Surface Area (m²)
Area (m²)
50000
40000
30000
20000
10000
0
50,308 m²
Air-Cooled requires
~18.7x larger area!
3,648 m²
Evaporative Condenser
Air-Cooled Finned-Tube
High Pressure P_high vs Vessel Limit
Condensing Pressure P_high (bar)
25
20
15
10
5
0
Vessel Design Limit (20 bar)
21.54 bar
15.64 bar
Evaporative Condenser
Air-Cooled Finned-Tube

4. Central Decanter Tank & Fresh Air Sensitivity

• **Central Water Decanter Sizing:** Governed by **Monsoon Day (89.7 m³/day = 3,738 kg/h)**. Differs from chiller sizing because it depends purely on latent dehumidification rate (m_da * delta_w). Pure condensate collected offsets **24.0% of evaporative cooling tower makeup water demand**.
• **Fresh Air Sensitivity (30% to 40%):** On Monsoon Night, outdoor air near saturation (28 °C, 90% RH) with low sensible loads causes indoor RH to drift to **76.2% at 30% fresh air and 76.6% at 40% fresh air**, violating the 75% ASHRAE comfort ceiling.
• **Engineering Remedy:** Subcooling-and-reheat: cooling coil operates at deep subcooling (ADP = 14.5 °C) followed by **2.8 K terminal reheat (410 kW thermal duty)** harvested from flue gas economizer / absorber heat, restoring room RH to 50.0%.
• **Single Biggest Design Lever:** Solution Heat Exchanger effectiveness (eps_SHX). Recirculation ratio is 3.5 to 4.5x refrigerant flow. Increasing eps_SHX from 0.70 to 0.80 saves 890 kW of generator heat, boosting COP by +10.0% and saving 6.5 kg/s of flue gas.

